

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Linq;
5 using System.Text;
6 using System.Threading.Tasks;
7 using CustomProjectRPG.Core.Utilities;
8 using CustomProjectRPG.Interfaces;
9 using CustomProjectRPG.Core;
10
11
12 namespace CustomProjectRPG.Entities
13 {
14
15     public abstract class Entity : ICollidable, IDraw, IUpdate
16     {
17         // Private fields for encapsulation.
18         private string _name;
19         private string _description;
20         private Vector2D _position;
21         private Hitbox _hitbox;
22
23         // Constructor for the Entity class.
24         // It initializes the name, description, position, and hitbox.
25         protected Entity(string name, string description, Vector2D startPosition, Vector2D hitboxSize)
26         {
27             _name = name;
28             _description = description;
29             _position = startPosition;
30             _hitbox = new Hitbox(startPosition, hitboxSize);
31         }
32
33         // Public properties for controlled access to the private fields.
34         public string Name
35         {
36             get { return _name; }
37         }
38
39         public string Description
40         {
41             get { return _description; }
42         }
43
44         public Vector2D Position
45         {
46             get { return _position; }
47             set
48             {
```

```
49         _position = value;
50         _hitbox.Position = value;
51     }
52 }
53
54 public Hitbox Hitbox
55 {
56     get { return _hitbox; }
57 }
58
59 // Abstract methods to be implemented by derived classes.
60 public virtual bool IsCollidingWith(ICollidable other)
61 {
62     return this.Hitbox.Intersects(other.Hitbox);
63 }
64
65 public abstract void OnCollision(ICollidable other);
66 public abstract void Draw();
67 public abstract void Update();
68
69
70 // Moves the entity to a new position.
71
72 public virtual void MoveTo(float x, float y)
73 {
74     this.Position = new Vector2D(x, y);
75 }
76
77 }
78
79 }
80
```